

SAT Memo 03 — The Redesign

VCE Algorithmics (HESS) · Unit 4 Outcome 2 · SAT Criteria 8–10

Issued Week 7 · Due Friday 4 September 2026 (Submission 3), with Amendments A05 & A06

The engineering situation

The board accepted your Memo 02 audit and has funded one redesign cycle. Your brief: improve the efficiency of the facility navigation solution — without losing fitness for purpose. Where the audit identified an intractable core, the board expects an engineered trade-off, not a promise of a faster exact algorithm.

What you must deliver

1. **Memo 03** — the design of an improved data model and algorithm combination, comprising:
 - an **improved design**: the new data model and algorithm combination, with the change from Memo 01 stated precisely (what was replaced, what was kept, and why the combination remains coherent); **C8**
 - **advanced technique(s) applied**: divide and conquer, dynamic programming, backtracking, and/or heuristic methods (greedy construction, hill climbing, random restarts, simulated annealing) — with the choice justified against alternatives; **C9**
 - a **time complexity analysis** of the improved solution (same standard as Memo 02: tools named, bounds justified, tally); **C9**
 - a **comparison to the original**: bounds side by side, quality trade-offs stated honestly (optimality guarantees surrendered, approximation behaviour observed), and a fitness-for-purpose conclusion against the original problem requirements. **C10**
2. **Completed workbook** (`marimo_memo03`) — design working, experiments, and comparison data.
3. **Amendment A05** and **Amendment A06** (below).

Amendments

A05 — The rush-hour constraint

C9

Head Office adds an operational constraint: during shift changes, up to m corridors are congested and must be avoided where possible, with the congestion set changing every few minutes. Your improved solution must respond to a new congestion set *without* full recomputation. Specify how your redesign accommodates this (incremental update, restricted recomputation, or heuristic repair), and analyse the per-update cost.

A06 — The optimality audit

C10

If your redesign uses a heuristic: design and run a small experiment comparing heuristic answers against exact answers on instances small enough to solve exactly, and report the observed quality gap. If your redesign is exact: demonstrate on a worked instance that the improved solution produces the same answers as the original at lower cost, and state the input scale at which the difference becomes material.

What is assessed

C8	Skills in the design of an improved data model and algorithm combination.
C9	Skills in advanced algorithmic problem-solving.
C10	Skills in the comparison of algorithmic solutions in terms of their coherence and fitness for purpose.

A short in-class **test** accompanies Submission 3 (advanced design techniques; date confirmed in class).

Timeline and support

Week 7, P2	Workshop 1 — bottleneck → technique selection (exit ticket)
Week 8, P2	Workshop 2 — improved design + analysis drafting
Thu 3 Sep	Checkpoint — workbook complete, declaration signed
Fri 4 Sep	Submission 3 due

Authentication requirements

As for Submissions 1 and 2: seed unchanged and on the cover sheet; substantive work in class workshops and your own workbook; methods may be discussed, but the redesign of *your* facility must be your own; authentication record signed at each checkpoint.

Marking emphasis

- **C8** rewards a *coherent* redesign — data model and algorithm still fit together and still solve the stated problem. A faster algorithm on a data model that no longer supports the problem’s queries fails coherence.
- **C9** rewards technique applied *with judgment*: the choice defended against at least one alternative, parameters (e.g. cooling schedule) chosen with reasons, limitations acknowledged.
- **C10** rewards honesty: what was gained (bound, measured time), what was surrendered (guarantees, exactness), and whether the trade serves the original purpose. “Faster but wrong for the use case” scores poorly; saying so about your own design scores well.